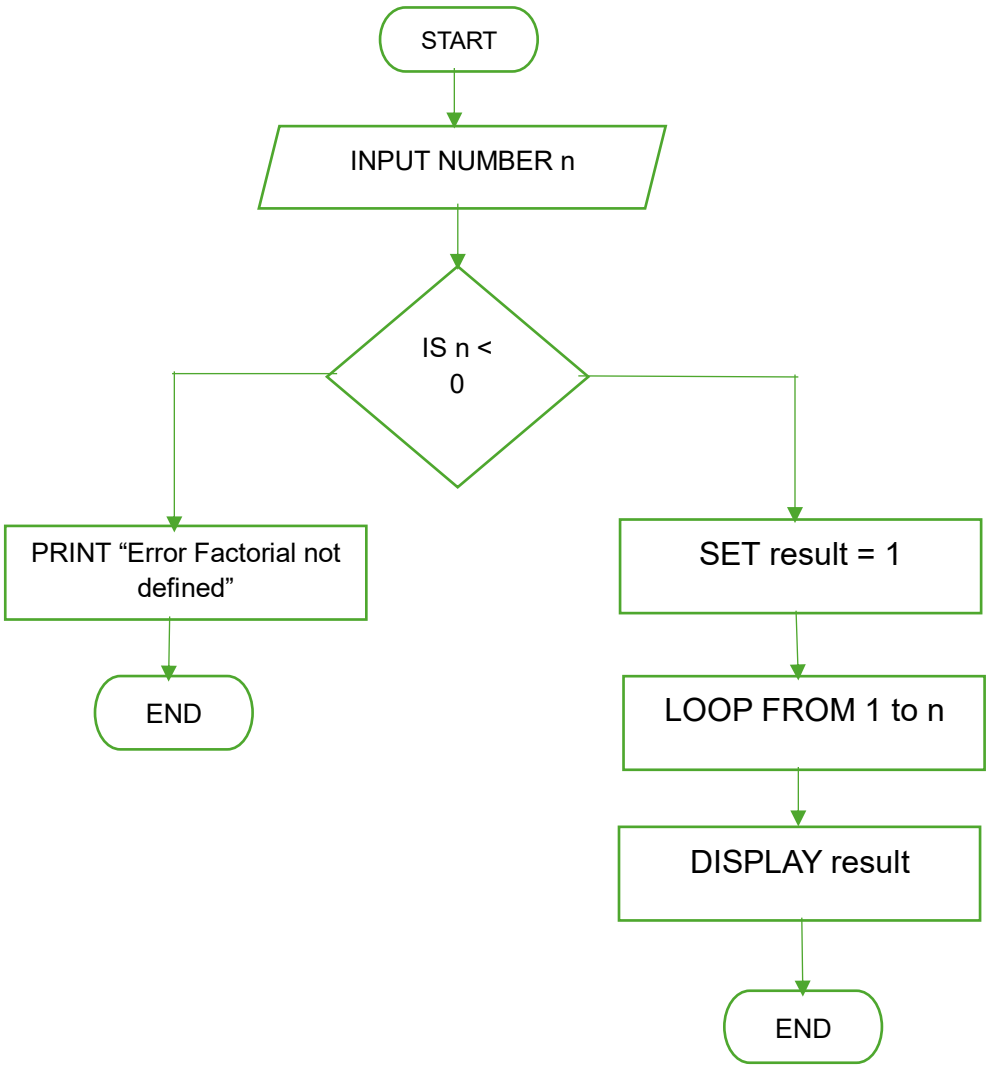
FACTORIAL

WORKFLOW



PSEUDO CODE

BEGIN

```
        INPUT n

        IF n < O THEN
            PRINT “Factorial not defined for negative numbers”
        ELSE
            SET result = 1

            FOR i = 1 TO n DO
                result = result * 1
            END FOR

            PRINT result
        END IF

    END
```

PYTHON CODE

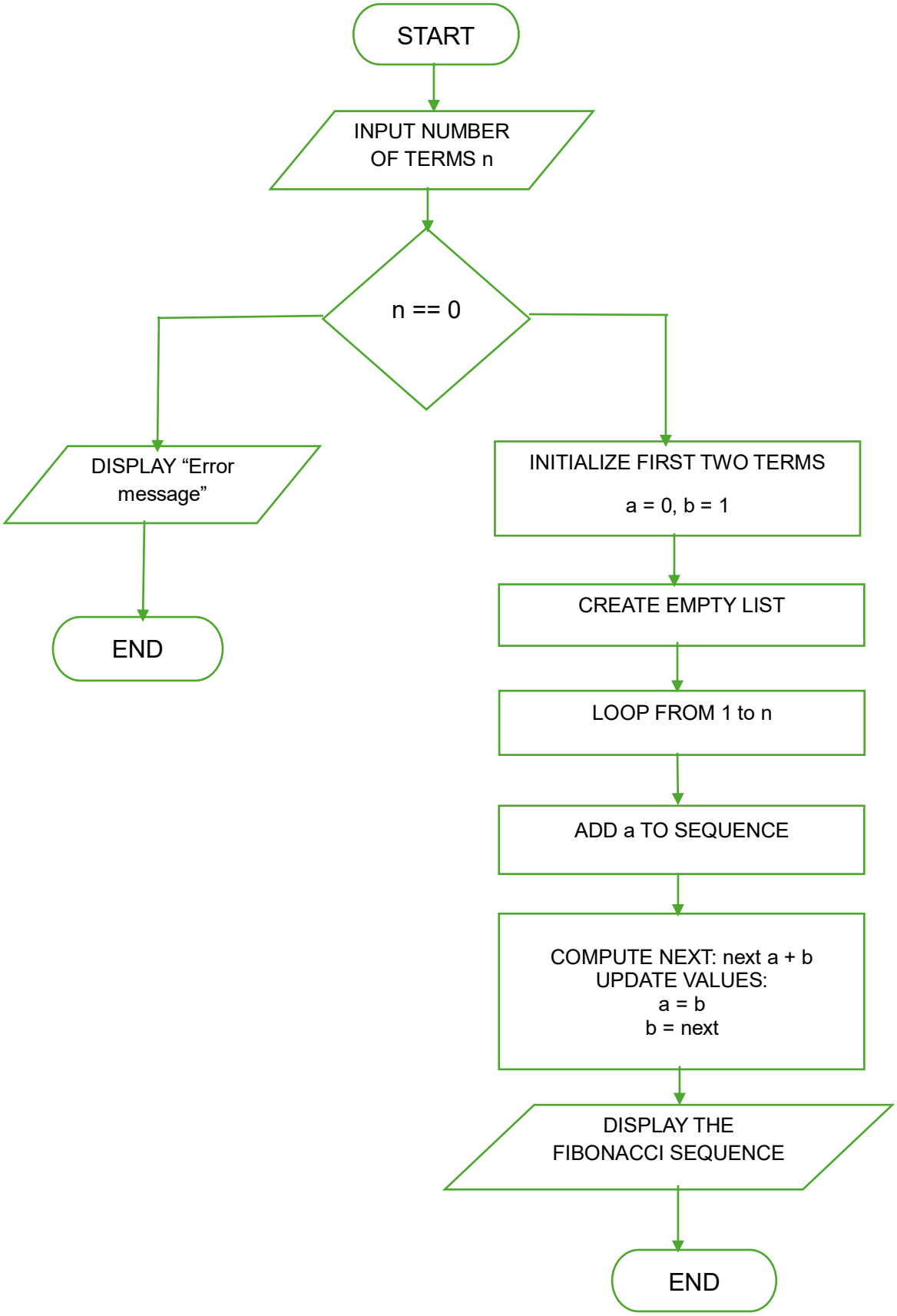
```
def factorial_with_steps(n):
    if n < 0:
```

```
        return None
    result = 1
    steps = ""
    for i in range(1, n + 1):
        result *= i
        steps += str(i)
        if i < n:
            steps += " × "
    return result, steps
while True:
    print("\nFactorial Calculator")

    user_input = input("Enter a number (or type 'exit' to quit): ")
    if user_input.lower() == 'exit':
        print("Goodbye!")
        break
    if not user_input.isdigit():
        print("Invalid input. Please enter a positive integer.")
        continue
    num = int(user_input)
    answer = factorial_with_steps(num)
    if answer is None:
        print("Factorial not defined for negative numbers.")
    else:
        result, steps = answer
        print(f"\nWorking:")
        print(f"{num}! = {steps}")
        print(f"{num}! = {result}")
```

FIBONACCI

WORK FLOW



PSEUDO CODE

```
BEGIN
   INPUT n

   IF n <= 0 THEN
      PRINT "Please enter a number greater than 0"
   ELSE
      SET a = 0
      SET b = 1
      CREATE empty list sequence

      FOR i = 1 TO n DO
         ADD a TO sequence

         SET next = a + b
         SET a = b
         SET b = next
      END FOR

      PRINT "Fibonacci Sequence:", sequence
   END IF
END
```

PYTHON CODE

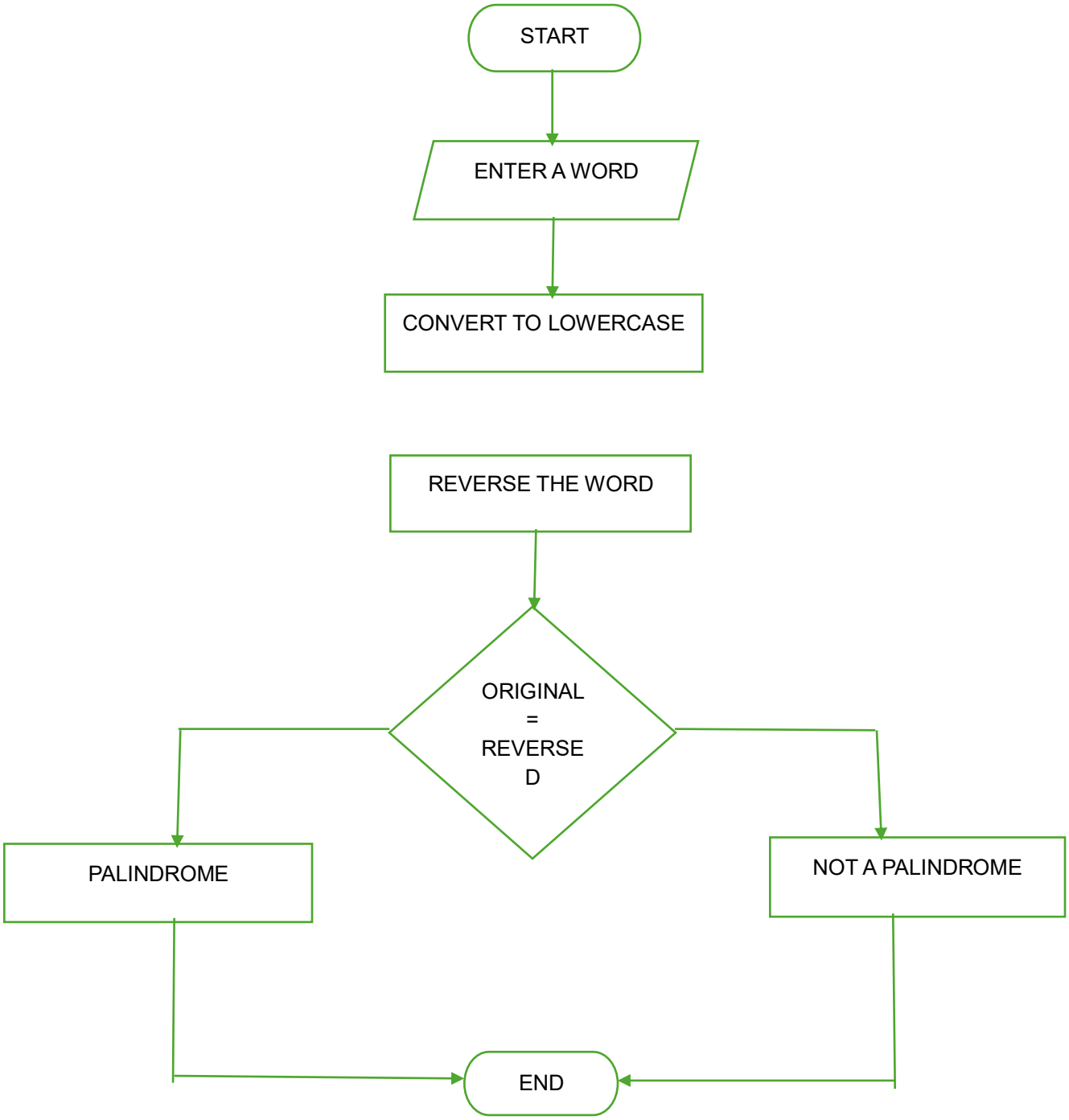
```
def fibonacci_with_steps(n):
    if n <= 0:
        return None
    sequence = []
    steps = []
    a, b = 0, 1
    for i in range(n):
```

```
        sequence.append(a)
        if i == 0:
            steps.append("0")
        elif i == 1:
            steps.append("1")
        else:
            steps.append(f"{a} = {sequence[i-2]} + {sequence[i-1]}")
        a, b = b, a + b
    return sequence, steps
while True:
    print("\nFibonacci Calculator")

    user_input = input("Enter how many terms (or type 'exit' to quit): ")
    if user_input.lower() == 'exit':
        print("Goodbye!")
        break
    if not user_input.isdigit():
        print("Invalid input. Please enter a positive integer.")
        continue
    num = int(user_input)
    result = fibonacci_with_steps(num)
    if result is None:
        print("Please enter a number greater than 0.")
    else:
        sequence, steps = result
        print("\nFibonacci Sequence:")
        print(sequence)
        print("\nWorking (Steps):")
        for step in steps:
            print(step)
```

PALINDROME

WORK FLOW



PSEUDO CODE

```
BEGIN
    PRINT “Enter a word”
    INPUT word

    SET word = convert word to lowercase
    SET reversed_word = reverse of word

    IF word == reversed_word THEN
        PRINT “The word is a Palindrome”
    ELSE
        PRINT “The word is NOT a Palindrome”
    END IF
END
```

PYTHON CODE

```
def check_palindrome():
```

```
    print("Palindrome Checker")
    word = input("Enter a word: ")
    # Step 1: Normalize
    normalized_word = word.lower()
    print(f"\nStep 1: Convert to lowercase → {normalized_word}")
    # Step 2: Reverse
    reversed_word = normalized_word[::-1]
    print(f"Step 2: Reverse the word → {reversed_word}")
    # Step 3: Compare
    print(f"Step 3: Compare '{normalized_word}' with '{reversed_word}'")
    if normalized_word == reversed_word:
        print("\nResult: ✅ It IS a palindrome!")
    else:
        print("\nResult: ❌ It is NOT a palindrome.")
# Run the program
check_palindrome()
```